we find that $PM = \sqrt{3}$ units.

Now, from the right-angled triangle PQM,

$$\sin 30^o = \frac{opposite\ side}{hypotenuse} = \frac{QM}{PQ} = \frac{1}{2}$$

$$\cos 30^o = \frac{adjacent\ side}{hypotenuse} = \frac{PM}{PQ} = \frac{\sqrt{3}}{2}$$

$$\tan 30^o = \frac{opposite\ side}{adjacent\ side} = \frac{QM}{PM} = \frac{1}{\sqrt{3}}$$

The reciprocals of these ratio can be easily foundout to be

$$\text{cosec}\, 30° = 2, \sec 30° = \frac{2}{\sqrt{3}}$$

and $\cot 30° = \sqrt{3}$

We will use the same triangle but the other angle of measure 60° now.

$$\sin 60^o = \frac{opposite\ side}{hypotenuse} = \frac{PM}{PQ} = \frac{\sqrt{3}}{2}$$

$$\cos 60^o = \frac{adjacent\ side}{hypotenuse} = \frac{QM}{PQ} = \frac{1}{2}$$

$$\tan 60^o = \frac{opposite\ side}{adjacent\ side} = \frac{PM}{QM} = \frac{\sqrt{3}}{1} = \sqrt{3}$$

The reciprocals of these ratio can be easily foundout to be

$$\text{cosec}\, 60° = \frac{2}{\sqrt{3}};\ \sec 60° = 2$$

and $\cot 60° = \frac{1}{\sqrt{3}}$

6.2.3 Trigonometric ratios of 0° and 90°

To find the trigonometric ratios of 0° and 90°, we take the help of what is known as a unit circle.

A unit circle is a circle of unit radius (that is of radius 1 unit), centred at the origin.

Why make a circle where the radius is 1unit?

This means that every reference triangle that we create here has a hypotenuse of 1unit, which makes it so much easier to compare angles and ratios.

We will be interested only in the positive values since we consider 'lengths' and it is hence enough to concentrate on the first quadrant.

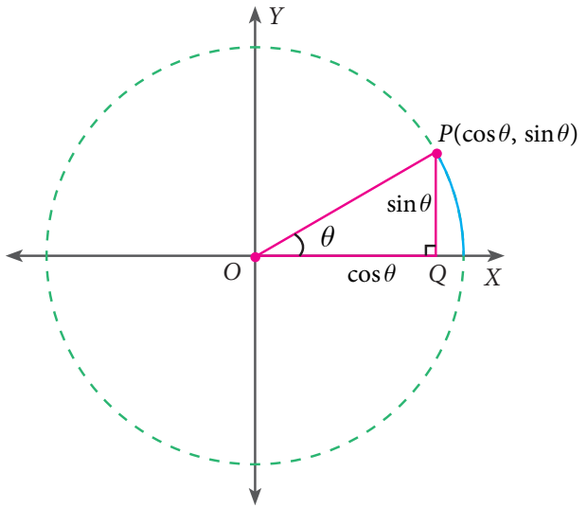


Fig. 6.11

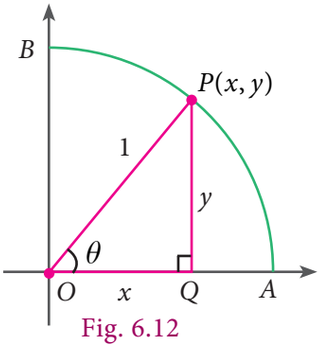


Fig. 6.12

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